

ME 4010

Tutorial 3

The objective of this tutorial is to get you comfortable with building observers and minimum order observers in Simulink.

Upload your PDF, simulink model and associated .m file by **midnight**.

1. For a double spring mass damper system with spring stiffnesses of $K_1 = 4 \text{ Nm}$ and $K_2 = 12 \text{ Nm}$, and damping of $C_1 = 2 \text{ Ns/m}$ and $C_2 = 5 \text{ Ns/m}$, and masses $M_1 = 3 \text{ Kg}$ and $M_2 = 5 \text{ KG}$,
 - a. Develop a controller in simulink to make the 1st mass oscillate with a frequency of 1 Hz and an amplitude of 0.8 m.
 - b. Add a minimum order observer to the design and compare your results.